Karan Lokchandani

Gandhinagar, Gujarat, India

+91 9975773675 karanlokchandani@gmail.com Linkedin Github

Education

Indian Institute of Information Technology Vadodara August 2023 – June 2027

B Tech in CSE *Gujarat*

Projects

AI Song Generator | *Python, Suno API, Groq, Laama 3.1, SwiftUI* | GitHub

- Developed an agentic AI-based song generation system consisting of three collaborative agents.
- Integrated Groq and Laama 3.1 models to conduct real-time research for contextual song generation.
- Implemented a lyrics generation agent powered by NLP models, producing high-quality, theme-aligned content.
- Integrated Suno AI's Bark model to convert generated lyrics into expressive vocals and music.
- Built a SwiftUI-based user-friendly frontend for seamless interaction and customization.
- **Impact**: Generated personalized songs, demonstrating real-time generative AI capabilities.

AI-Driven Video Lecture Generator | *Python, OpenAI and Gemini API, Mermaid, FastAPI* | GitHub

- Designed a generative AI platform to convert PDF books into engaging video lectures with AI-generated teacher voices.
- Enhanced user experience with dynamic slide generation and real-time educational roadmap creation using Mermaid.
- Utilized GPT-based models to enable interactive feedback and contextual adjustments for lecture content.
- Planned future development to include virtual teacher avatars for immersive learning.
- **Impact**: Improved accessibility and customization in online education, boosting engagement by 50%.

Ventus – File Synchronization System | *Rust, Swift (SwiftUI), Uni-FFI, Python, C* | GitHub

- Implemented real-time file synchronization with an event-driven architecture using gRPC for low-latency updates.
- Developed a robust multi-platform system, supporting macOS, iOS, Linux, and Windows with seamless integration.
- Optimized performance by reducing sync latency to under 50ms and implementing an efficient custom FTP protocol.
- Utilized AI-based conflict resolution strategies to enhance collaboration during simultaneous edits.
- **Stats**: 4 platforms supported, 50ms latency, 200 MB/s transfer speeds under ideal conditions.

KVStore-DB | *Rust, Bitcask, CLI, Unit Testing* | GitHub

- Engineered a high-performance key-value storage solution based on the Bitcask storage model.
- Achieved millisecond-level read/write latencies by optimizing disk I/O with write-ahead logging techniques.
- Developed CLI tools for efficient data querying and management, with support for large-scale datasets.
- **Stats**: Supported over 1M keys with consistent sub-1ms operation times under high loads.

Technical Skills

Programming Languages: Rust, Swift, C++, Python, JavaScript
Frameworks & Libraries: Actix, LangChain, Streamlit, Flask, FastAPI, SwiftUI
Tools & Platforms: Git, Docker, gRPC, Postman, VS Code, Xcode, IntelliJ IDEA, Arch Linux, Bash, PowerShell
Databases: PostgreSQL, SQLite, Redis
Core Competencies: Event-driven architecture, API design and integration, real-time systems, distributed systems, and blockchain development.
Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Object-Oriented Programming.

Achievements

- Specialist on Codeforces (Max: 1419)
- 3-star on CodeChef (Max: 1751)
- Runner-Up in the Open Innovation Track of IIIT Vadodara's flagship Hackathon, HackIIITV
- Two-time Winner of the "Tech Hunt" CTF organized by IIIT Vadodara's Tech Fest, Cerebro